

Who Does the Recommender Serve? A Fairness Audit and Mitigation of Demographic Bias in Distributed Rating Classification on MovieLens 1M

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Abstract—Rating-prediction models are routinely evaluated on aggregate accuracy, which hides whether a model serves different demographic groups equally. We audit a distributed logistic-regression classifier that predicts whether a user rates a movie highly (rating ≥ 4) on the MovieLens 1M dataset, implemented as an Apache Spark ML pipeline. We first identify and correct a target-leakage flaw in a common feature recipe—user- and movie-average ratings computed over the full dataset—which modestly inflated accuracy from 0.723 to a corrected 0.719; correcting it is nonetheless a prerequisite for an honest audit. On the corrected model, aggregate accuracy of 0.72 conceals sharply different behavior by protected attribute: gender disparity is mild (demographic parity difference, DPD, of 0.036; disparate impact, DI, of 0.95), but age disparity is substantial (DPD = 0.146, DI = 0.80, at the common four-fifths threshold), with the model under-selecting items for younger users and over-selecting for older users. We then evaluate two mitigations. Removing the protected attributes from the feature set (“fairness through unawareness”) did not help and worsened the age gap (DPD 0.146 $\rightarrow$ 0.215), because correlated proxies remain. Per-group threshold calibration nearly eliminated the gender gap (DPD $\rightarrow$ 0.001) at a negligible accuracy cost (0.0008), but, being attribute-specific, left the age gap unchanged. Our results argue that fairness must be measured per attribute, that dropping protected features is insufficient, and that leakage-aware evaluation is a precondition for trustworthy fairness claims.

Index Terms—algorithmic fairness, recommender systems, Apache Spark, logistic regression, demographic bias, MovieLens, disparate impact

I. INTRODUCTION

Recommender and rating-prediction systems increasingly mediate what content users see, yet their evaluation is dominated by aggregate accuracy that treats all users as interchangeable. A model can report strong overall accuracy while systematically under-serving a demographic subgroup. As machine learning moves into consequential settings, this gap between aggregate performance and per-group performance has become central to trustworthy AI [1], [2].

We study this on a concrete, reproducible artifact: a logistic-regression classifier predicting the binary target “high rating” (rating ≥ 4) for a (user, movie) pair on MovieLens 1M [3], built as an Apache Spark ML pipeline [4], [5]. The setting

is deliberately mainstream—logistic regression on MovieLens is a canonical baseline—which makes it a useful lens on a broader question: *when a standard, well-performing pipeline is deployed at scale, who does it actually serve?*

Three findings organize the paper. First, the pipeline as originally constructed suffered from *target leakage*: user- and movie-average rating features were computed over the entire dataset, including the ratings being predicted. Correcting this changes accuracy only modestly (0.723 $\rightarrow$ 0.719), but it is a precondition for an honest audit. Second, the corrected model’s aggregate accuracy of 0.72 conceals a strong asymmetry between protected attributes: gender disparity is mild, whereas age disparity is large and sits at the four-fifths disparate-impact threshold. Third, the intuitive mitigation of removing protected attributes fails and even worsens the age gap, while a cheap per-group threshold calibration removes the gender gap almost entirely but does not transfer across attributes.

Contributions.

- We identify and correct a target-leakage flaw in a common MovieLens feature recipe and quantify its (modest) inflationary effect (§III-C, §V).
- We conduct a group-fairness audit of a scalable Spark logistic-regression classifier across gender and age using DPD, equal-opportunity difference (EOD), and disparate impact, showing that a single aggregate score hides an attribute-dependent disparity (§V).
- We evaluate two interpretable mitigations and characterize the fairness–accuracy tradeoff, including a negative result for feature removal (§VI).

II. RELATED WORK

Fairness definitions. Dwork et al. introduced fairness through awareness [6]; Hardt et al. proposed equalized odds and equal opportunity, framing fairness as error-rate parity across groups [7]; Feldman et al. formalized disparate impact and its removal [8]. Verma and Rubin catalog these often-conflicting definitions [9]. We report demographic parity, equal opportunity, and disparate impact together rather than committing to one.

Bias in recommendation. Ekstrand et al. show recommender effectiveness and evaluation can differ across demographic groups [10], and surveys document how bias enters at data, model, and evaluation stages [1]. Most such work targets ranking or top- k recommendation; we audit the upstream rating-classification stage, where disparities originate but are rarely measured.

Mitigation. Mitigations are pre-, in-, or post-processing. Kamiran and Calders study pre-processing via reweighing [11]; post-processing adjusts decision thresholds per group [7]. We evaluate one pre-processing approach (removing sensitive attributes) and one post-processing approach (per-group thresholds), both interpretable and compatible with a distributed pipeline. Our feature-removal result is a concrete instance of the well-known limitation of “fairness through unawareness” [6].

Gap we address. Prior fairness studies rarely (i) operate on a distributed big-data engine, (ii) tie the audit to a reproducible interpretable model, and (iii) control for evaluation artifacts such as target leakage. We address all three.

III. METHODOLOGY

A. Dataset

MovieLens 1M [3] contains 1,000,209 ratings from 6,040 users on $\sim 3,900$ movies, with per-user gender, age band, and occupation. We load the “::”-delimited files with explicit schemas, drop duplicate (user, movie) ratings, enforce referential integrity, and restrict ratings to [1, 5].

B. Task and features

The target is $y = \mathbb{I}[\text{rating} \geq 4]$. Features are (i) *demographic* (age, occupation, gender), (ii) *static movie* attributes (release year, movie age relative to 2000, number of genres), and (iii) *derived interaction* statistics (user-average rating, movie-average rating, user rating count, movie popularity). Gender and age serve both as baseline features and as the *protected attributes* for the audit.

C. Leakage-aware feature construction

The derived interaction statistics are label-dependent: a user’s average rating over all their ratings includes the rating being predicted. Computing them over the full dataset before splitting leaks the test label into test features. We correct this by performing the 80/20 split *first* (seed 42) and computing all derived statistics on the training split only, joining them onto test rows and backing off unseen users/movies to the global training mean (3.582). We also build a leaky reference model (statistics over all data) to measure the effect.

D. Model

The classifier is a Spark ML Pipeline of `VectorAssembler`→`StandardScaler`→`LogisticRegression` (max 100 iterations). Logistic regression is chosen for interpretability: coefficients expose which features drive predictions.

TABLE I
EFFECT OF LEAKAGE CORRECTION (TEST SET, $n = 199,705$).

Model	Accuracy	Precision	Recall	F1
Leaky (full-data averages)	0.723	0.731	0.821	0.773
Corrected (train-only)	0.719	0.727	0.817	0.770

E. Fairness metrics

For a protected group g , let the selection rate be $s_g = P(\hat{y} = 1 \mid g)$ and the true-positive rate $t_g = P(\hat{y} = 1 \mid y = 1, g)$. We report

$$\text{DPD} = \max_g s_g - \min_g s_g, \quad \text{DI} = \min_g s_g / \max_g s_g, \quad (1)$$

$$\text{EOD} = \max_g t_g - \min_g t_g. \quad (2)$$

DPD and DI capture demographic parity; EOD captures equal opportunity. A DI below 0.8 is a common disparate-impact threshold [8].

F. Mitigations

Sensitive-feature removal retrain with gender and age excluded. *Per-group threshold calibration* keeps the baseline model but selects a per-gender decision threshold so each group’s selection rate matches the overall target rate—post-processing toward demographic parity.

IV. EXPERIMENTAL SETUP

Experiments run on a single machine in Spark local mode (`local[*]`), Spark 3.5.3, Java 17, Python 3.11. The split uses seed 42; the training set has 800,504 rows and the test set 199,705. The primary accuracy metric is overall accuracy; the primary fairness metric is gender DPD, complemented by age DPD, EOD, and DI. We audit gender (two groups) and age (seven MovieLens bands). Reported F1 is for the positive class.

V. FAIRNESS AUDIT

A. Leakage correction

Table I reports the leaky and corrected models. Correcting the leakage moved accuracy from 0.723 to 0.719 and positive-class F1 from 0.773 to 0.770. The effect is modest—each average aggregates ~ 165 ratings per user, so removing one rating barely shifts it—but the correction is methodologically necessary and all subsequent results use the corrected model.

B. Group disparities

Table II shows per-gender behavior. The gender gap is mild: DPD = 0.036, DI = 0.95 (well above the 0.8 threshold), and EOD = 0.007—by these measures the model is close to parity across gender. The picture differs sharply for age. As Fig. 1 shows, the baseline selection rate rises almost monotonically with user age, from 0.59 for the 18–24 band to 0.73 for the 56+ band, giving an age DPD of 0.146, DI of 0.80 (at the four-fifths threshold), and EOD of 0.090. Aggregate accuracy of 0.72 therefore masks a substantial age disparity while gender looks benign.

TABLE II
GENDER FAIRNESS OF THE CORRECTED BASELINE.

Group	n	Sel. rate	TPR	Accuracy
Female	49,177	0.673	0.823	0.710
Male	150,528	0.637	0.816	0.722
Gender: DPD = 0.036, EOD = 0.007, DI = 0.95				
Age: DPD = 0.146, EOD = 0.090, DI = 0.80				

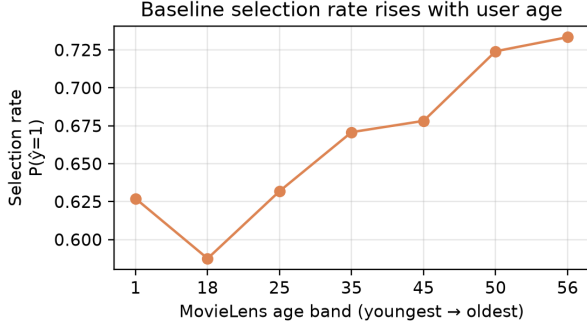


Fig. 1. Baseline selection rate $P(\hat{y} = 1)$ rises with user age band, revealing a disparity invisible in the aggregate accuracy of 0.72.

VI. MITIGATION AND TRADEOFF

Table III compares the corrected baseline with the two mitigations on gender, and Fig. 2 shows the DPD for both protected attributes across all three configurations.

Feature removal fails. Dropping gender and age from the feature set did not improve gender parity (DPD 0.036 $\rightarrow$ 0.042) and *worsened* the age gap (DPD 0.146 $\rightarrow$ 0.215, DI 0.80 $\rightarrow$ 0.72). Removing protected attributes leaves correlated proxies—movie age, genre mix, popularity—through which the model reconstructs the same disparity, a concrete instance of the limits of fairness through unawareness [6].

Per-group thresholds work, but only for the targeted attribute. Calibrating a per-gender threshold (male 0.488, female 0.522) reduced gender DPD from 0.036 to 0.001 (DI = 0.999) at an accuracy cost of just 0.0008 (0.719 $\rightarrow$ 0.718). Because the thresholds are gender-specific, the age gap was unchanged (DPD = 0.146). This illustrates that post-processing is cheap and effective but must be applied to each attribute one intends to protect. Fig. 3 plots the gender fairness–accuracy tradeoff: per-group thresholding moves left (fairer) with essentially no vertical (accuracy) drop.

VII. DISCUSSION, LIMITATIONS, AND THREATS TO VALIDITY

Interpretation. The audit yields three transferable lessons: (i) a single aggregate metric can be simultaneously acceptable and misleading—here gender looks fair while age does not; (ii) removing protected attributes does not remove bias and can worsen it via proxies; and (iii) cheap post-processing closes a targeted gap but does not generalize across attributes.

Internal validity. Results use a single split and seed. We fix seed 42 for reproducibility but do not report cross-seed vari-

TABLE III
FAIRNESS–ACCURACY COMPARISON ON GENDER.

Configuration	Accuracy	Gender DPD	Gender DI
Corrected baseline	0.719	0.036	0.95
Sensitive-feature removal	0.719	0.042	0.94
Per-group thresholds	0.718	0.001	1.00

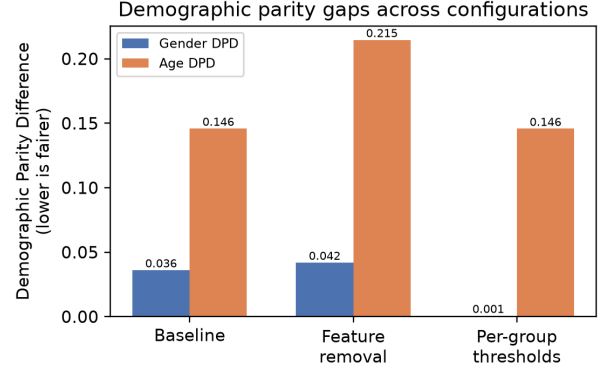


Fig. 2. Demographic parity difference by protected attribute across configurations. Feature removal worsens the age gap; per-group (gender) thresholds close the gender gap but not the age gap.

ance or significance tests; adding them is the most important next step before strong claims. Per-group thresholding also assumes group membership is available at inference, which raises governance questions of its own.

External validity. MovieLens users are not a representative population, and gender is a self-reported binary; the audit cannot speak to non-binary users or to production traffic, and results should not be generalized without re-auditing.

Construct validity. Demographic parity, equal opportunity, and disparate impact can conflict [9]; we report all three and do not claim the mitigated model is “fair” in an absolute sense—only that specific measured gaps change as reported.

VIII. CONCLUSION AND FUTURE WORK

We audited a scalable, interpretable rating classifier on MovieLens 1M for demographic fairness. After correcting a target-leakage flaw, we found a mild gender disparity but a substantial age disparity hidden beneath a 0.72 aggregate accuracy; removing protected attributes failed to help and worsened the age gap, while per-group thresholding closed the gender gap almost for free but did not transfer to age. Future work includes cross-seed significance testing, stronger baselines (gradient-boosted trees), age-aware and multi-attribute thresholding, in-processing fairness constraints, and extending the audit from rating classification to top- k recommendation.

REPRODUCIBILITY

All numbers are produced by a single script (`fairness_study.py`) from the raw MovieLens 1M files, with seed 42, Spark 3.5.3, Java 17, and Python 3.11.

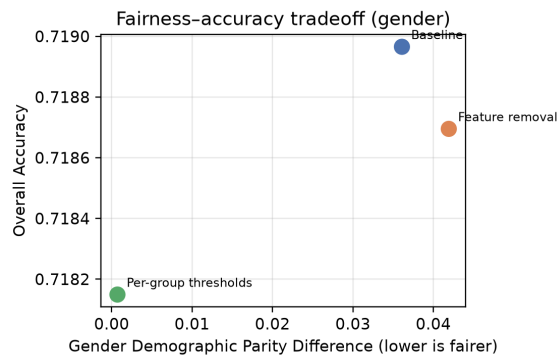


Fig. 3. Gender fairness-accuracy tradeoff. Per-group thresholding reaches near-zero DPD at negligible accuracy cost.

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